

A PROPOSAL FOR THE TELECOM FOUNDATION EDUCATION SYSTEM

# The notebook that **teaches back.**

A QR code on every subject notebook turns a finished chapter into a quiz, a rank, and a merit bursary — across all 16 TEFOS schools.

Chapter quizzes

Merit bursaries

Inter-school championship

💡 THE IDEA IN ONE BREATH

- 01 A student finishes Chapter 1.
- 02 They scan the book they are already holding.
- 03 They answer four questions.
- 04 They earn points toward next month's bursary.



**No app.**  
**No signup.**  
**No form.**  
**No behaviour change.**

— which is precisely why it will work.

## THE INSIGHT

# The QR code is **already** in the book

This is the only marketing surface a school owns that the student touches voluntarily, every day.

**It meets the student at peak attention**

They have just finished the chapter. The book is open. The page is right there.

**It requires zero new behaviour**

No app to download, no code to memorise, no website to visit, no reminder to send.

**It travels home and gets photographed**

It sits open on a desk where visitors see it, and ends up in photos, on desks, in parents' houses.

## Every other school channel is a push.

-  Newsletters get ignored.
-  Flyers get binned.
-  Apps get deleted.

✦ This one is a pull.

## THE STUDENT JOURNEY

# Five steps. About **ninety seconds**.

**01****Scan**

The QR on the cover of any subject notebook.

**02****Enter code**

Six characters printed inside the front cover.

**03****Pick chapter**

The one they just finished. Tap Go.

**04****Answer**

Five questions, mostly about their own class.

**05****Rank up**

Points, a rank tier, and a place on the board.



**There is no step six.** That is the entire design. Everything in this proposal exists to keep students at step five.



 THE PARTICIPATION ENGINE

# Five non-negotiable rules

Break any one of these and the programme dies quietly within a term.

1

## Points are never negative

A student who tries and gets it wrong must never end up worse off than one who never scanned.



2

## Participation earns the first rank

A student who takes part always has an identity, even at zero percent.



3

## Only a first attempt moves the board

Retries pay a little, so nobody is locked out — but nobody can farm the board.



4

## Show growth, not distance

“You gained 180 points”, not “you are 2,400 short” — unless the gap is genuinely reachable.



5

## Never publish only the top

Champions, most improved, and participation rate. Always all three.



## THE ONE NUMBER THAT MATTERS MOST

# Punishing a wrong answer kills participation

The arithmetic below is the single most important line in this proposal.

**✗ AS ORIGINALLY SPECIFIED****+10 correct · -5 wrong**

|               |            |
|---------------|------------|
| 5 of 5 right  | <b>+50</b> |
| 3 of 5 right  | <b>+5</b>  |
| 2 of 5 right  | <b>-5</b>  |
| All wrong     | <b>-25</b> |
| Never scanned | <b>0</b>   |

**✓ AS BUILT****+10 · 0 · +3 retry · +25 perfect**

|               |            |
|---------------|------------|
| All right     | <b>+75</b> |
| Half right    | <b>+25</b> |
| Two of five   | <b>0</b>   |
| All wrong     | <b>0</b>   |
| Never scanned | <b>0</b>   |



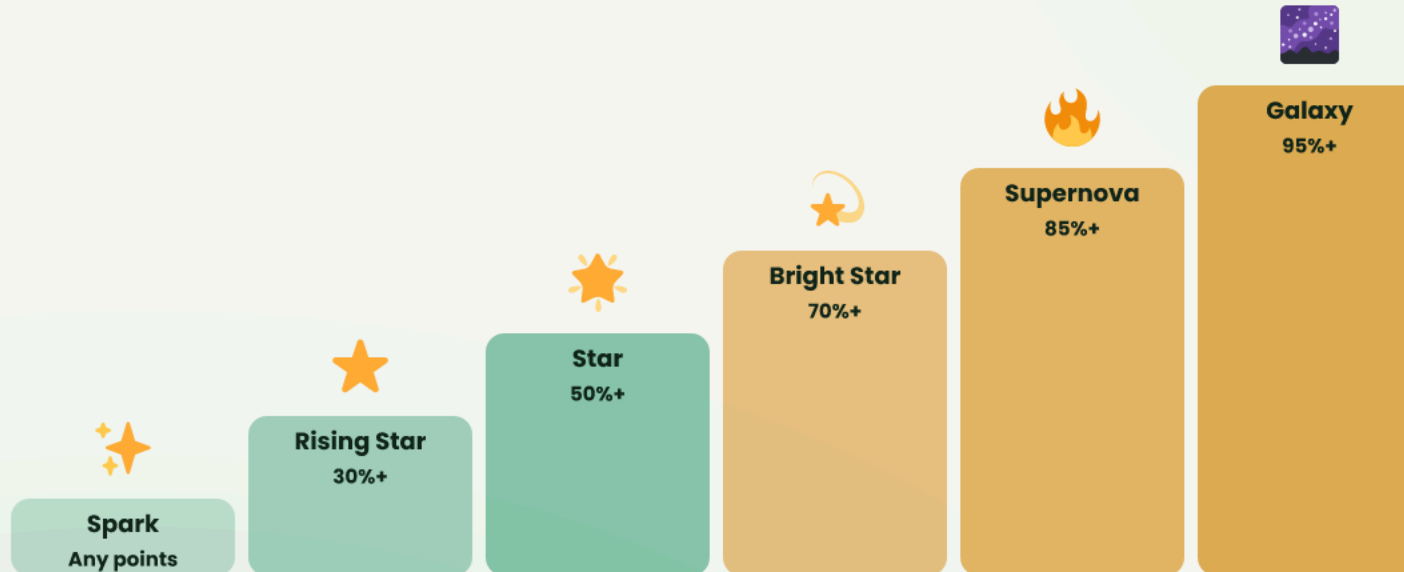
**With negatives, a student who scans and does badly is worse off than one who ignores the book entirely. The programme punishes exactly the behaviour it needs most.**



## ★ RANK LADDER

# Spark to Galaxy

Every student has an identity. Nobody sits in a dead zone below the first rung.



Rank = points earned ÷ points available. Tier appears after 5 chapters attempted.



## THE ACTIVITY FLOOR

Rank is points earned ÷ points available — never a raw average — and only appears after 5 chapters attempted.

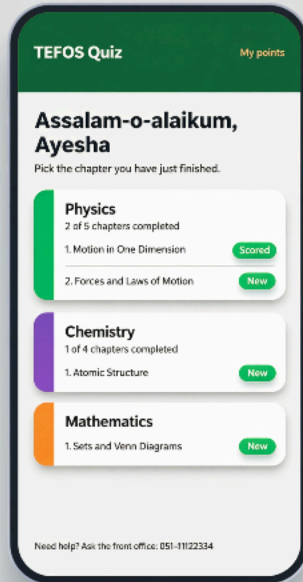
Without the floor, a student sits one easy quiz, scores 95% and quits — and out-ranks the student who showed up every week.



**Bug we found and fixed:** the floor applied to the data but not to the screen — a one-quiz student still read “Galaxy” on their own result. Now the displayed tier is clamped too.

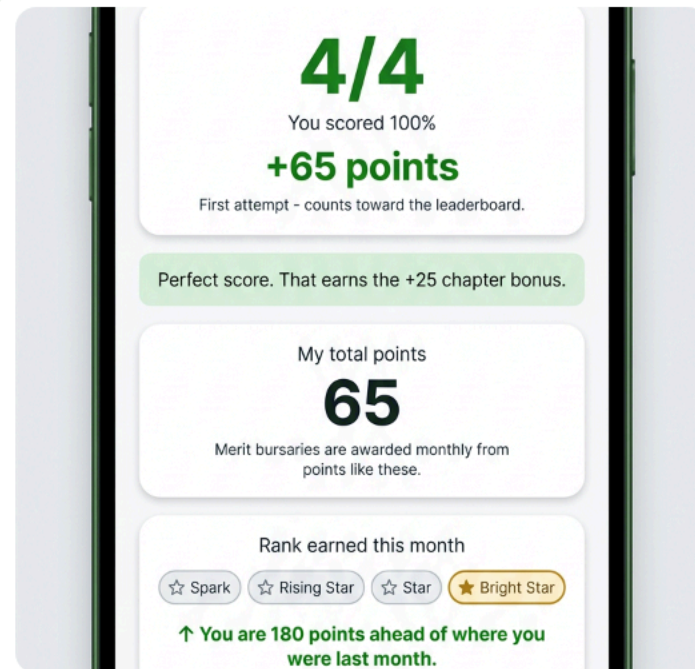
## WHAT THE STUDENT SEES

# Three screens, ninety seconds



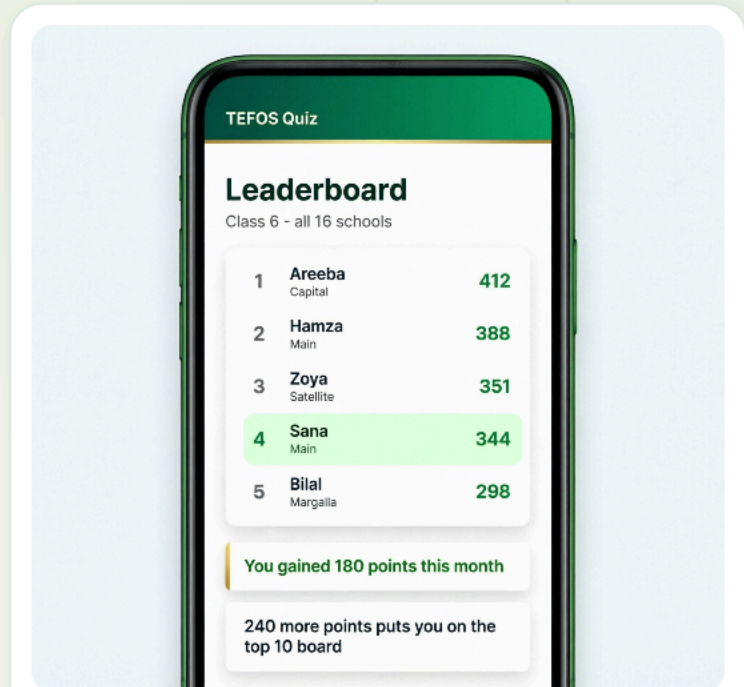
## The chapter hub

Every subject, every chapter, one tap to start.



## The payoff

Points, rank tier, and how far they have come this month.



## The 16-school board

All campuses together — plus the reach line, shown only when close.

## INTEGRITY

# A quiz nobody can google

Half the questions are about things that happened in that specific classroom, that week.



## THE USUAL KIND

**"A body travels 120 m in 15 s at constant speed. What is its speed?"**

Searchable in four seconds. The prize goes to whoever has the fastest internet connection and the most willing relatives.



## THE KIND WE USE

**"On the first day of the Motion unit, which classroom activity did we do?"**

**"Our lab report was due on which date?"**

Nobody can search these. Only a student who was there can answer — we are paying for attention, not for guessing.

**0**

answer keys sent to any device

**5**

chapters before a rank shows

**1**

first attempt per chapter, enforced

**2.5 KB**

per quiz page, gzipped



## MOTIVATION

# Show growth. Show the gap **only** when it is reachable.

The most common mistake in school gamification is telling a child exactly how far they are from winning.

 ALWAYS

**"You are 180 points ahead of where you were last month."**

Progress against yourself. This is what sustains effort.

 ONLY IF REACHABLE

**"240 more points puts you on the top 10 board."**

A real nudge, shown only to students within striking distance.

 NEVER

**"You are 2,400 points short of the board."**

The gap is too big to close, so the rational move is to stop trying.



**Publish three boards, always.**



**Champions**

Top scorers



**Most improved**

Biggest monthly gain



**Participation rate**

How many took part

 THE ECONOMICS

# The school pays in tuition credit, **not cash**

This is the trick that makes the whole programme affordable at scale.

 WHAT IT IS

## Merit bursary

Up to 10 bursaries a month, each worth up to 10% of one month's tuition fee, awarded on points earned.

Cost to the school: foregone revenue on a student who already pays. Budgetable, capped, predictable.

 WHAT IT MUST NOT BE CALLED

## "Students earn their own fee."

Identical budget. But it replaces a student election with a leaderboard, invites fee-regulation scrutiny, and reads badly in a newspaper.

Same money, completely different posture.



**The honest caveat:** a bursary helps a wealthy family more than a poor one, because a rich family can absorb the fee. Keep a small, separately-funded need-based component — and let sponsors underwrite that, because it is the strongest education-access outcome report available.

## 17 RHYTHM

# Three cadences, three different jobs

A yearly reward attached to a daily habit is how engagement dies around week six.

## WEEKLY



## Star of the Week

Recognition only. House points, a name on the school board, and nothing else. Theatre, and that is fine — as long as nobody expects cash.

## MONTHLY



## The merit bursary

The real reward. Fee credit, published, capped, and actually paid. This is the one that parents care about.

## EVERY 2 MONTHS



## The inter-school championship

The big one. Sixteen campuses, trophies, a ceremony, and a camera. This is what sponsors pay for.



**The monthly rhythm is what keeps the notebook open.** The championship is the photo opportunity. Neither replaces the other.

 THE BIG EVENT

# Sixteen school champions, not five finalists

A live online final for 48 finalists is how you end up with a cheating scandal and a dead programme.



## PHASE 1 · SCHOOL HEAT

### Two weeks, asynchronous, in the app.

Every student in the class can enter. Same questions, done in their own time inside a two-week window. No timing pressure, no dropped connections, no proctoring — and participation stays near total.



## PHASE 2 · CITY FINAL

### Sixteen students. One afternoon.

One champion per school. Live, but sixteen people, all on campus, on good connections, with staff in the room. Genuinely manageable — and it makes a real ceremony worth filming.

| Place                                                                                                             | Reward                                                             | How many |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------|
|  <b>School champion</b>          | Bursary + trophy + name on the school banner + a slot in the final | 16       |
|  <b>City winner 1–3</b>        | Laptop or tablet                                                   | 3        |
|  <b>City 4–5</b>               | Fee waiver                                                         | 2        |
|  <b>Top school of the term</b> | Inter-school trophy and banner                                     | 1        |



Sixteen champions means sixteen proud schools — and sixteen ceremonies' worth of photographs.

## THE PRIZE LADDER

# Rewarding the middle, not just the peak

The top 20% are already motivated by grades. The programme lives or dies in the middle.

| Band                                                                                                            | Reward                                               | Cadence | Why                                                                          |
|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------|---------|------------------------------------------------------------------------------|
|  <b>Anyone taking part</b>     | Badge, name on the monthly board, a branded bookmark | Monthly | Participation is the goal, so reward the act of participating                |
|  <b>50%+ (the middle half)</b> | Merit bursary — real money off the fee               | Monthly | The largest group, the biggest family impact, easiest for sponsors to fund   |
|  <b>70%+</b>                   | Certificate and merit gear                           | Monthly | Recognition, not material                                                    |
|  <b>Top 1 per year group</b>   | Laptop                                               | Annual  | A real trophy — eight winners, not one, which removes the incentive to cheat |
|  <b>House champion</b>       | Trophy at the ceremony                               | Annual  | The photo opportunity, and the sponsor's logo                                |



**Removed:** “a branded pen for 40%.” It tells a child they did badly and hands them a consolation prize. That is worse than giving nothing.



Eight laptops, not one. Spreading the top prize across year groups is what makes cheating pointless.

## THE MARKETING ENGINE

# Every printed batch carries its own **tracked link**

This is how the school answers the only question that matters: which notebooks actually worked?



**`https://<host>/t/{batch}/{subject}`**

→ redirects to the code entry screen, and records the scan



## One link per print run

Main campus Physics vs Capital campus Maths are separately attributable from day one.



## Measured, not guessed

Scans, participation and average score per batch, per campus, per subject — the real marketing number.



## Zero cost

The tracking is a line of code. There is no extra spend to attribute it.



## AND THE BIGGEST LEVER

# Zero-rated data

Students are on metered mobile data. If the operator zero-rates this host, participation does not improve slightly — it changes order of magnitude.

A full quiz page already costs 2.5 KB gzipped, so this is a cheap ask with an enormous ceiling. It is the single most valuable thing an affiliated operator can contribute.



## SPONSORSHIP

# Sixteen schools **changes the conversation**

Stop selling “logo on a quiz”. Sell the championship.

## ✗ WHAT DOES NOT SELL

### “Put your logo on our quiz.”

This is a marketing sponsorship. Education CSR budgets are earmarked for scholarships, infrastructure and teaching — a completely different budget line at most corporates.

## ✓ WHAT SELLS

### “Title sponsor of a 16-school championship.”

16 campuses, 2,000+ students, a recurring streamed event, trophies, media coverage, and a clean education-access outcome story. That is a headline CSR asset.

| Tier                                                                                                      | What they buy                                                                   | How easy  |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------|
|  <b>Scholarship fund</b> | “We underwrite N merit bursaries” — maps perfectly to education-access outcomes | Easiest   |
|  <b>In-kind prizes</b> | Printers, tablets, ISP kit, local vouchers — hugely valuable, almost no cash    | Very easy |
|  <b>Named prize</b>    | “This month’s top scorer wins a [brand] kit”                                    | Moderate  |
|  <b>Presented by</b>   | Title sponsor of the biennial championship ceremony                             | Moderate  |

 PRIVACY AND SAFEGUARDING

# Built GDPR-K grade, by default

Pakistan has no enacted statute comparable to GDPR or COPPA. That is not a licence to collect loosely.



## Nothing personal is ever typed in

No signup form. No name, no father's name, no class, no roll number. The student enters a 6-character printed code.



## Nothing identifying is ever shown

Public boards show first name and campus only. No guardian names, no student numbers, no photographs.



## Every point is traceable

The points ledger is append-only. A parent can audit every single point their child has ever been awarded.



## A child can always reach an adult

The front office number is printed inside the front cover, permanently, regardless of what the server is doing.



**Before launch:** obtain parental consent at enrolment, publish a plain-language privacy notice inside the front cover, and confirm ICT Directorate guidance with local counsel.



## EXECUTION

# Four phases, and a **kill switch** at every step



## PHASE 1

**4–6 weeks**

One subject, one chapter, one class. No sponsors, no ceremony, no bursary. The only question: will they scan?



## PHASE 2

**Months 2–3**

All subjects. Weekly house boards. First merit bursary actually paid out. Teacher dashboard switched on.



## PHASE 3

**Months 4–6**

First inter-school championship. Ceremonies, trophies, and the first sponsorship conversation.



## PHASE 4

**Year 2**

Championship becomes a fixture. Sponsor renews. New campus cohorts onboarded.



**The kill switch.** If monthly participation is below 50% at the end of phase 1, stop and fix it. Do not scale a weak programme across 16 campuses — that is how a good idea dies of bad management rather than bad design.



Phase 1 needs one volunteer science teacher — not the most eager head of department.

## MEASUREMENT

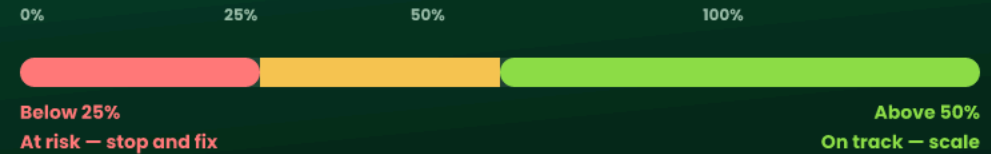
# One number decides whether this lives

Track it weekly. Everything else is diagnostic.



## MONTHLY PARTICIPATION

Share of enrolled students who opened at least one chapter this month



| What we also watch                                                                                                         | Why it matters                                                  | Who acts on it   |
|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|------------------|
|  <b>Participation rate per campus</b>     | Which campuses are engaged, and which need a teacher's nudge    | Campus principal |
|  <b>Scans per printed batch</b>           | Which notebook print run actually worked — the marketing number | Marketing / HQ   |
|  <b>Average score per chapter</b>         | Which chapters the cohort actually struggles with               | Subject teacher  |
|  <b>Share of students reaching 50%+</b> | Whether the programme is lifting the middle, not just the top   | Head of school   |
|  <b>Points redeemed as bursary</b>      | Actual cost to the school, against the 0.5–1% cap               | Finance          |



The teacher-facing report is not a by-product. It is the reason teachers support the programme instead of quietly resenting it.

## WHAT IT COSTS

# Almost nothing to start

| Item                                                                                                                           | Cost                  | Note                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------|
|  <b>QR code printed on the notebook cover</b> | ~PKR 0                | You are already printing these. This is a design change.              |
|  <b>Access code inside the front cover</b>    | ~PKR 0                | Printed by the school alongside the class list.                       |
|  <b>The application</b>                       | PKR 0                 | No licences, no hosting minimum, no vendor. Runs on one small server. |
|  <b>Deployment to 16 campuses</b>             | Minimal               | Server-side software update, not a campus-by-campus installation.     |
|  <b>Merit bursary pool</b>                    | 0.5–1% of fee revenue | Capped, published, and the real cost. Budget it deliberately.         |
|  <b>Inter-school championship</b>             | Per-event             | Sponsorship-funded. Do not run phase 1 on sponsor money.              |

**0**

new vendors or licences

**16**

campuses from one server update

**≤1%**

of fee revenue — the entire real cost

 WHAT WE NEED

# Three decisions

Nothing here needs a new budget line. All three are policy choices.

## 01



### Name a pilot

One class, one subject, one campus. Six weeks. A science teacher who volunteers, not the most eager head of department.

## 02



### Agree the framing

“Merit bursary”, never “earn your own fee”. And quiz points never decide Head Boy, monitors or prefects — that stays exactly as it is today.

## 03



### Open the operator conversation

Zero-rated data on the quiz host, and a first look at sponsoring the inter-school championship. Both are worth more than the software.



Total cost to pilot: the price of six weeks of one teacher's goodwill.



All three can be decided in this meeting.

# It is not a quiz app. It is a reason to open the notebook.

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Every other school initiative asks a student to do something extra. This one pays them for what they were already doing — which is why it will still be running in March.